Guillen Ramirez, Hugo Armando

PhD in Computer Science

PERSONAL DETAILS

Date of birth: 21 November 1989 · Nationality: Mexican · One child

Researcher unique identifier(s):

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EDUCATION AND KEY QUALIFICATIONS

28/11/2018	**PhD in Computer Science**, Department of Computer Science, Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE), Mexico. Supervisor: Dr Israel Marck Martínez-Pérez. Thesis: Methods for non-coding RNA gene prediction.
2014	**MSc in Computer Science**, CICESE, Mexico. Supervisor: Dr Israel Marck Martínez-Pérez. Thesis: A Tissue P-System and Molecular Algorithm for the MAX-CLIQUE Problem.
2011	**BSc in Computer Systems Engineering**, Instituto Tecnológico de Tepic, Mexico. Award for the highest grade in the 2007–2011 graduating class.

CURRENT POSITIONS

10/2026 – present	**Maître d'enseignement et de recherche (Senior Lecturer and Researcher), 50%**, Section of Medicine, Faculty of Science and Medicine, Université de Fribourg, Switzerland. Teaching and research profile in clinical real-world AI and medical informatics.
03/2023 – 03/2027 (fixed term)	**Advanced Postdoctoral Researcher, 100% until 09/2026, 50% since 10/2026,** Department of Visceral Surgery and Medicine, Inselspital, Bern University Hospital and University of Bern, Switzerland. Supervisor: Prof. Guido Beldi. Also affiliated with the Department for BioMedical Research and, since 2026, the Department of Digital Medicine.

PREVIOUS POSITIONS

2021 – 2023 **Marie Skłodowska-Curie Fellow**, School of Biology and Environmental Science, University College Dublin, Ireland. Mentor: Prof. Rory Johnson. Project RNADOMAIN.

2019 – 2021 **Postdoctoral Researcher**, Department for BioMedical Research, University of Bern, Switzerland. Mentor: Prof. Rory Johnson. Computational genomics of long noncoding RNA.

2019 **Postdoctoral Researcher**, CICESE, Mexico. Mentor: Dr Salvador Villarreal-Reyes. Artificial intelligence for public health.

GRANTS AND AWARDS

Digitalization Commission of the University of Bern (DigiK UniBE), 2025–2029. Secure International Collaboration: Advancing Medical AI with Federated Learning and Synthetic Data. CHF 500,000. As named associated researcher I conceived and lead the AI and methods component: federated training and evaluation of CARESCORE across Inselspital and CHUV Lausanne, and synthetic perioperative data. PI of record: Prof. G. Beldi; co-applicant: Dr O. Endrich.

Marie Skłodowska-Curie Individual Fellowship (H2020-MSCA-IF-2019, project 884178), 2021–2023, EUR 196,591. Principal investigator and sole fellow; project RNADOMAIN, University College Dublin, Ireland. Europe-wide competitive fellowship.

CONACYT National Scholarship, Mexico, 2011–2018. Support for master's and doctoral studies.

ARTICLES IN PEER-REVIEWED JOURNALS

2026

Guillen-Ramirez, H., Lucas, K. L., Wintsch, Y. M., Blatter, T. U., Triep, K., Endrich, O., & Beldi, G. (2026). Subgroup deployment-readiness audit of calibration and clinical utility in perioperative infection prediction. Journal of the American Medical Informatics Association. In press. https://doi.org/10.1093/jamia/ocag184

Blatter, T. U., Wintsch, Y., Triep, K., Endrich, O., **Guillen-Ramirez, H.*,** & Beldi, G. (2026). End-of-surgery prediction of postoperative infectious complications from intraoperative vital-sign dynamics. *npj Digital Medicine*. **[*Shared senior authorship]** https://doi.org/10.1038/s41746-026-02707-1

Guillen-Ramirez, H., … , Beldi, G., & Endrich, O. (2026). LLM-augmented semantic embeddings enable Cross-Lingual mapping of medical procedure terms. *Scientific Reports*. https://doi.org/10.1038/s41598-025-34778-7

2025

Guillen-Ramirez, H., Sanchez-Taltavull, D., Perrodin, S., Peisl, S., Triep, K., Gaudet-Blavignac, C., ... & Beldi, G. (2025). Prediction of postoperative infections by strategic data imputation and explainable

machine learning. Journal of the American Medical Informatics Association, 32(11), 1706-1717. https://doi.org/10.1093/jamia/ocaf145

Zwicky, S. N., Spari, D., Rodjakovic, D., **Guillen-Ramirez, H.**, Yilmaz, B., & Beldi, G. (2025). Preoperative Enterosignatures Predict Surgical Site Infections After Abdominal Surgery. In Open Forum Infectious Diseases (Vol. 12, No. 9, p. ofaf549). US: Oxford University Press. https://doi.org/10.1093/ofid/ofaf549

2024

Peisl, S., Sánchez-Taltavull, D., **Guillen-Ramirez, H.**, Tschan, F., Semmer, N. K., Hübner, M., ... & Keller, S. (2024). Noise in the operating room coincides with surgical difficulty. BJS open, 8(5), zrae098. https://doi.org/10.1093/bjsopen/zrae098

Peisl, S., **Guillen-Ramirez, H.***, Sánchez-Taltavull, D., Widmer, A., Sommerstein, R., & Beldi, G. (2024). Influence of patient characteristics on microbial composition in surgical-site infections: insights from national surveillance study. British Journal of Surgery, 111(6), znae138. https://doi.org/10.1093/bjs/znae138 **[*Shared first authorship]**

Dhaka, B., Zimmerli, M., Hanhart, D., Moser, M. B., **Guillen-Ramirez, H.**, Mishra, S., ... & Johnson, R. (2024). Functional identification of cis-regulatory long noncoding RNAs at controlled false discovery rates. Nucleic Acids Research, 52(6), 2821-2835. https://doi.org/10.1093/nar/gkae075

2023 and earlier

Esposito, R., …, **Guillen-Ramirez, H.,** ... & Johnson, R. (2023). Tumour mutations in long noncoding RNAs enhance cell fitness. Nature communications, 14(1), 3342. https://doi.org/10.1038/s41467-023-39160-7

Esposito, R., …, **Guillen-Ramirez, H.A.**, …, & Johnson, R. (2022). Multi-hallmark long noncoding RNA maps reveal non-small cell lung cancer vulnerabilities. Cell genomics, 2(9), p.100171. https://doi.org/10.1016/j.xgen.2022.100171

Bosch-Guiteras, N., Uroda, T., **Guillen-Ramirez, H.A.**, Riedo, R., Gazdhar, A., Esposito, R., Pulido-Quetglas, C., Zimmer, Y., Medová, M. & Johnson, R. (2021). Enhancing CRISPR deletion via pharmacological delay of DNA-PKcs. Genome Research. https://doi.org/10.1101/gr.265736.120

Aguilera-Mendoza, L., Marrero-Ponce, Y., García-Jacas, C. R., Chavez, E., Beltran, J. A., **Guillen-Ramirez, H. A.**, & Brizuela, C. A. (2020). Automatic construction of molecular similarity networks for visual graph mining in chemical space of bioactive peptides: an unsupervised learning approach. Scientific reports, 10(1), 1-23. https://doi.org/10.1038/s41598-020-75029-1

Calvillo-Vázquez, J. G., **Guillén-Ramírez, H. A.**, DiazDuarte-Rodríguez, M., Licea-Claverie, A., & Méndez, E. R. (2019). Particle size distribution from extinction and absorption data of metallic nanoparticles. Applied optics, 58(36), 9955-9966. https://doi.org/10.1364/AO.58.009955

Aguilera-Mendoza, L., Marrero-Ponce, Y., Beltran, J. A., Tellez Ibarra, R., **Guillen-Ramirez, H. A.**, & Brizuela, C. A. (2019). "Graph-based data integration from bioactive peptide databases of pharmaceutical interest: toward an organized collection enabling visual network analysis." Bioinformatics. https://doi.org/10.1093/bioinformatics/btz260

Guillen-Ramirez, H. A., & Martínez-Pérez, I. M. (2018). "Classification of riboswitch sequences using k-mer frequencies". Biosystems, 174, 63-76. https://doi.org/10.1016/j.biosystems.2018.09.001

ARTICLES IN PROCEEDINGS

2026: Lucas, K. L., Wintsch, Y. M., Blatter, T., Triep, K., Endrich, O., **Guillen-Ramirez, H. A.**, & Beldi, G. (2026). Two Actionable Windows: Disentangling Early Mortality From Late Infection Risks Using Time-Resolved AI. BJS, 113(Supplement_5).

2018: Ibarra-Hernández, R. F., Bertin, N., Alonso-Arévalo, M. A., & **Guillén-Ramírez, H. A.** (2018). "A benchmark of heart sound classification systems based on sparse decompositions". In Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series (Vol. 10975). https://doi.org/10.1117/12.2506758

2017: Guillen-Ramirez, H. A., Colbes, J., Brizuela, C. A., & Martinez-Perez, I. M. (2017). "Accurate classification of immunomodulatory RNA sequences". In 2017 International Joint Conference on Neural Networks (IJCNN) (pp. 236-241). IEEE. https://doi.org/10.1109/IJCNN.2017.7965860

2013: Brizuela, C. A., Luna-Taylor, J. E., Martinez-Perez, I., **Guillen, H. A.,** Rodriguez, D. O., & Beltran-Verdugo, A. (2013). "Improving an evolutionary multi-objective algorithm for the biclustering of gene expression data". In 2013 IEEE Congress on Evolutionary Computation (pp. 221-228). IEEE. https://doi.org/10.1109/CEC.2013.6557574

TEACHING

2026

- *Introduction to Image Analysis* — Biomedical Engineering Master's Program, Artificial Intelligence in Medicine Master's Program and Bioinformatics Master's Program, University of Bern, Switzerland. **Graduate level**.
- *Programming for Biologists (Introduction to Python)* — Faculty of Medicine, University of Bern, Switzerland. **Graduate level**.
- *Human Medicine, 3rd year medical students* — Faculty of Medicine, University of Bern, Switzerland. **Graduate level**.
- *Introduction to Precision Medicine* — Pharmaceutical Master's Program, Bioinformatics Master's Program, University of Bern, Switzerland. **Graduate level**.
- *Clinical Decision Support (Human–Computer Interaction for CDSS)* — Artificial Intelligence in Medicine Master's Program, ARTORG Center, University of Bern, Switzerland. **Graduate level.**
- *Introduction to R* — Faculty of Medicine, University of Bern, Switzerland. **Graduate level**.
- *Medical Informatics* — Biomedical Engineering Master's Program, University of Bern, Switzerland. **Graduate level**.

2025

- *Introduction to R* — Faculty of Medicine, University of Bern, Switzerland. **Graduate level**.
- *Introduction to Clinics* — Artificial Intelligence in Medicine Master's Program, University of Bern, Switzerland. **Graduate level**.
- *Medical Informatics* — Biomedical Engineering Master's Program, University of Bern, Switzerland. **Graduate level**.

2024

- *Introduction to Clinics* — Artificial Intelligence in Medicine Master's Program, University of Bern, Switzerland. **Graduate level**.

2023

- *Introduction to Clinics* — Artificial Intelligence in Medicine Master's Program, University of Bern, Switzerland. **Graduate level**.

2022

- *The RNA World* — University College Dublin, Ireland. **Graduate level**.

2019

- *Cancer Genomics* — University of Bern, Switzerland. **Graduate level**.
- *Computational Biology (Genome Indexing Algorithms)* — CICESE Research Center, Mexico. **Graduate level**.

2017–2019

- *Artificial Intelligence*, *Advanced Mathematics for Engineering*, *Evolutionary Computing*, and *Assembly Language* — Autonomous University of Baja California, Mexico. **Undergraduate level**.

2017–2018

- *Mathematics II, IV, V*, *Intermediate Algebra I & II*, and *Mathematical Reasoning V & VI* — Xochicalco High School, Mexico. **High school level**.

THESIS SUPERVISION

2021 – 2022: Co-supervised two master's theses: **Mateusz Brodzik** (Universitat Pompeu Fabra (UPF), Spain) and **Jules Montuelle** (Université de Fribourg, Switzerland).

MANUSCRIPTS UNDER REVIEW

Guillen-Ramirez, H., Endrich, O., & Beldi, G. (2026). Verification Debt as Agentic Artificial Intelligence Accelerates Clinical Analysis. Viewpoint, submitted to JMIR AI, 13 October 2026.

PREPRINTS

Guillen-Ramirez H, Endrich O, Beldi G. Verification Debt as Agentic Artificial Intelligence Accelerates Clinical Analysis (2026). JMIR Preprints. 13/10/2026:114328. https://doi.org/10.2196/preprints.114328

Guillen-Ramirez, H., Lucas, K. L., Wintsch, Y. M., Blatter, T. U., Triep, K., Endrich, O., & Beldi, G. (2026). Deployment-readiness audit of calibration, clinical utility, and fairness in perioperative infection prediction. medRxiv. https://doi.org/10.64898/2026.06.15.26355656 Accepted in the Journal of the American Medical Informatics Association (2026).

Guillen-Ramirez, H., Triep, K., Gaudet-Blavignac, C., Phull, B., Beldi, G., & Endrich, O. (2025). MAP-CARE: Enhancing Cross-Lingual Medical Intervention Terms Analysis Through LLM-supported Semantic Embeddings. https://doi.org/10.21203/rs.3.rs-6848278/v1 Published as Scientific Reports (2026) https://doi.org/10.1038/s41598-025-34778-7

CONFERENCE PRESENTATIONS AND POSTERS

Poster: *Intraoperative Vital Sign Variability as Early Warning Signals of Postoperative Infection.*
Blatter T., Wintsch Y., Triep K., Endrich O., Guillen-Ramirez H.A., Beldi G.
MCID Annual Event 2025, Switzerland.

Poster: *MAP-CARE: Advancing International Healthcare Through Enhanced Multilingual Medical Terminology Analysis.* Guillen-Ramirez H., Triep K., Beldi G., Endrich O.
Precision Health Day 2025, USI–SUPSI, Switzerland.

Talk: *Implementing Semantic Search for Cohort Selection Using Procedure-Based CHOP Codes.*
Guillen-Ramirez H., Triep K., Endrich O., Beldi G.
Annual Meeting of SGG-SGVC-SASL & SVEP 2024, Switzerland.

Talk: *Prediction of Postsurgical Infection by Explainable AI and Strategic Data Imputation.*
Guillen-Ramirez H., Sanchez-Taltavull D., Peisl S., Perrodin S., Triep K., Endrich O., Beldi G.
Annual Meeting of SGG-SGVC-SASL & SVEP 2024, Switzerland.

Talk: *Early Prediction of Postsurgical Infections Using a Bayesian Approach.*
Guillen-Ramirez H., Sanchez-Taltavull D., Peisl S., Perrodin S., Triep K., Endrich O., Beldi G.
Annual Meeting of SGG-SGVC-SASL & SVEP 2023, Switzerland.

Talk: *Age Is an Independent Predictor for the Shift From Skin to Intestinal Microbes in Surgical Site Infections Across Different Types of Procedures.*
Peisl S., Guillen-Ramirez H., Sanchez-Taltavull D., Sommerstein R., Beldi G.
Annual Meeting of SGG-SGVC-SASL & SVEP 2023, Switzerland.

Poster: *GeneHTracker: Improving Reproducibility and Reusability of Datasets Based on Gene Identifiers.*
Guillen-Ramirez H.A., Johnson R.
21st European Conference on Computational Biology (ECCB 2022), Sitges, Barcelona.

Poster: *Transposable Elements as Functional Domains of Long Noncoding RNAs.*
Guillen-Ramirez H.A., Johnson R.
2020 Virtual Meeting on Transposable Elements, Cold Spring Harbor Laboratory.

Talk: *Accurate Classification of Immunomodulatory RNA Sequences.*
Guillen-Ramirez H.A., Colbes J., Brizuela C.A., Martinez-Perez I.M.
International Joint Conference on Neural Networks (IJCNN 2017), Anchorage, Alaska, USA.

Poster: *Simulation of a Programmable Molecular Computer.*
Guillen H.A., Martinez-Perez I.
21st International Conference on DNA Computing and Molecular Programming (DNA 2015), Boston, USA.

Talk: *Challenges in the Prediction of Non-Coding RNA Genes* (translated from Spanish).
Guillen H.A., Martinez-Perez I.
National Meeting on Computer Science (Encuentro Nacional de Ciencias de la Computación, ENC 2014), Mexico.

Poster: *Design of a Tissue P-System for Solving the MAX-CLIQUE Problem* (translated from Spanish).
Guillen H.A., Martinez-Perez I.
6th International Congress on Computer Science (CiComp 2013), Mexico.

INVITED TALKS

2026, Feb “Representing Time Series, Missingness, and Semantics in Surgical Risk Prediction” in the workshop “AI Across Scales: From Molecules to Planet Earth” at the Isaac Newton Institute, University of Cambridge, UK

2020, Oct “Discrete mathematics and the design of CRISPR-Cas9 genome editing libraries” at Instituto Tecnológico de Tepic, Mexico.

2017, Mar “Score functions for discriminant k-mers and their application in ncRNA classification” (translated from Spanish) at CICESE Research Center, Mexico.

2016, Oct “Methods for ncRNA gene prediction: RNA secondary structure alternative graph representations” (translated from Spanish) in the XII National Meeting on Mathematical Biology (Center of Mathematical Sciences, UNAM Campus Morelia), Mexico.
2016, Oct “Bio-computing: RNA, BioPython, and Graphs” (translated from Spanish) in the Computer Science Fall School at CICESE Research Center, Mexico.

ATTENDED COURSES

2025 Supervising Master's Theses and Doctoral Dissertations. University of Bern, Switzerland.

2025 From Idea to Impact: Crafting Successful Grant Proposals. University of Bern, Switzerland.

2025 Missing Data and Imputation Methods (Swiss Institute of Bioinformatics), Lausanne, Switzerland.

2025 Causal Inference (Swiss Institute of Bioinformatics), Basel, Switzerland.

2025 Federated Learning in Bioinformatics (Swiss Institute of Bioinformatics), Lugano, Switzerland.

2023 Bayesian Statistics with R (Swiss Institute of Bioinformatics), Basel, Switzerland.

2023 How to illustrate your research results: Creating advanced vector graphics in Adobe Illustrator CC. University of Bern, Switzerland.

2022 EMBO Lab Leadership Postdoc Course (EMBO Laboratory Management Courses), streamed from Heidelberg, Germany.

2022 Networking For Postdocs - A Strategic Approach (University College Dublin), Ireland. 2022 Promoting Yourself And Your Research (University College Dublin), Ireland.

2022 Preparing Academic Applications And Interviews For Postdocs (University College Dublin), Ireland.

2022 Get That Paper Written And Published (University College Dublin), Ireland.

2022 Online Interview Skills For Postdocs (University College Dublin), Ireland.

2021 Single-cell Transcriptomics (Swiss Institute of Bioinformatics), streamed from Lausanne, Switzerland.

2021 Reproducible research and data analysis using Nextflow pipelines (Swiss Institute of Bioinformatics), streamed from Lausanne and Barcelona.

2021 Data Management Plan (Swiss Institute of Bioinformatics), streamed from Lausanne, Switzerland.

2020 Enrichment Analysis, (Swiss Institute of Bioinformatics), streamed from Lausanne, Switzerland.

2020 Advanced statistics: statistical modelling, (Swiss Institute of Bioinformatics), Lausanne, Switzerland.

2016 XVIII Fall School on Mathematical Biology (Center of Mathematical Sciences, UNAM Campus Morelia), Michoacán, Mexico.

2016 NGS Data Analysis and Informatics Conference, San Diego, USA.

2015 Applications of Artificial Organic Networks (National Meeting on Computer Science 2015), Baja California, Mexico.

MEMBERSHIP

Department of Digital Medicine, University of Bern (2026-Present)

OHDSI Switzerland - National Node (2025-Present)